Numer. Math. Theor. Meth. Appl.
doi: 10.4208/nmtma.OA-2025-0007

Vol. **18**, No. 4, pp. 947-971
November 2025

Numerical Analysis of a Second Order Crank-Nicolson Mixed Finite Element Method for the Swift-Hohenberg Equation

Hongling Li[1], Manting Xie[2] and Meiling Yue[3,*]

[1] *Department of Mathematics and Physics, Suqian University, Suqian 223800, China*
[2] *School of Mathematics and KL-AAGDM, Tianjin University, Tianjin 300350, China*
[3] *School of Mathematics and Statistics, Beijing Technology and Business University, Beijing 100048, China*

Received 14 January 2025; Accepted (in revised version) 6 May 2025

Abstract. In this paper, we conduct an error analysis for a mixed finite element method combined with Crank-Nicolson time-stepping for the Swift-Hohenberg (SH) equation. The original fourth-order differential equation from the SH equation is reformulated as a coupled system comprising a nonlinear parabolic equation and an elliptic equation. This transformation allows us to propose a mixed finite element method that utilizes only continuous elements for the approximation of the system. We rigorously demonstrate that our proposed scheme is unconditionally uniquely solvable and maintains unconditional nonlinear energy stability. Additionally, we provide a thorough analysis to determine the convergence rate of the method. To substantiate our theoretical findings, we present numerical tests that confirm both the accuracy and energy stability of the scheme.

AMS subject classifications: 35Q99, 65N30, 65M12, 65M70

Key words: Swift-Hohenberg equation, error analysis, mixed finite element, Crank-Nicolson unconditionally energy stable.

1. Introduction

Swift-Hohenberg equation is a nonlinear high-order evolution equation, which was first proposed in 1977 by Swift and Hohenberg [23] when describing Rayleigh Benard convection, that is, to describe the process of the evolution of a specific model in the fluid layer confined to the heat conduction boundary. Because of its complex and in-

*Corresponding author. *Email addresses:* yuemeiling@lsec.cc.ac.cn (M. Yue), lhl1210@126.com (H. Li), mtxie@tju.edu.cn (M. Xie)

 ©2025 Global-Science Press

teresting evolution process, it has become one of the important examples of nonlinear dynamic evolution leading to the formation of complex patterns. In addition to the study of complex fluids, it has also been applied in other fields, including the study of neural tissue [9], the simulation of laser problems [12] and the solution of the Kelvin problem on the minimum surface [6].

The SH equation (2.2) can be derived from a variation of Lyapunov functional (2.1), so the continuous equation has nonlinear stability [5, 23]. The equation includes high-order differential terms, nonlinear terms, and some physical parameters. The different selection of parameters makes the SH equation show a banded structure or hexagonal structure [4]. In recent years, various numerical methods for simulating the SH equation (2.2) have also been proposed. In [29], the authors presented a large time-stepping numerical method for the Swift-Hohenberg equation by adding an extra artificial term to preserve the unconditional energy stability. The famous scalar auxiliary variable (SAV) approach [22] has been widely used for the Swift-Hohenberg equation, see e.g., [18, 24, 25]. In [16, 17, 20], authors use various invariant energy quadratization (IEQ) schemes for the temporal discretization that also preserve the unconditional energy stability. High-order time-accurate (BDF2/BDF3) method has been adopted in [15, 19, 27]. For the spatial discretization, the finite difference methods [8, 14, 26], finite element method [19, 20, 24], discontinuous Galerkin (DG) methods [16], local discontinuous Galerkin method (LDG) [7, 30] and Fourier spectral methods [11, 28], have been widely reported.

It is worth noting that the Swift-Hohenberg equation (2.2) exhibits three critical features: high nonlinearity, small parameters ($0 < \epsilon \ll 1$), and high-order derivatives. In practical applications, these characteristics make it challenging to approximate the biharmonic terms using standard methods, potentially leading to more stringent requirements for time step sizes. To address this challenge, an effective approach is to split the biharmonic operator into two Laplacian operators, as demonstrated in [19–21]. In this paper, we propose a new fully discrete finite element scheme for the SH equation using a mixed formulation based on the Crank-Nicolson (CN) scheme, which ensures unconditionally unique solvability. We also establish the unconditional energy stability of this scheme. By employing the operator splitting method to handle the biharmonic term in (2.2), we use the mixed finite element method for spatial discretization and the CN time-stepping scheme for temporal discretization. Our primary focus is on providing a rigorous error analysis for the proposed numerical scheme. Furthermore, the presence of the biharmonic term complicates the application of conforming finite element methods, as it necessitates the use of C^1 elements. The mixed finite element method, however, allows us to utilize standard C^0 elements, which are supported by many computational software packages. Numerical results are provided to validate both the energy stability and accuracy of our numerical scheme. Additionally, we obtain and confirm the convergence properties through numerical experiments.

The paper is organized as follows. In Section 2, we introduce the Swift-Hohenberg equation and derive its mixed weak formulation. Section 3 focuses on the spatial discretization using a mixed finite element method. We establish the unconditional energy

stability of this approach and provide an optimal error estimate when it is coupled with the mixed FEM for spatial discretization. The fully discrete scheme, including its unconditional energy stability, is detailed in Section 4. Section 5 is dedicated to deriving error estimates for the fully discrete scheme. Various numerical experiments are conducted in Section 6 to evaluate the performance and validate the theoretical findings of the proposed fully-discrete scheme. Finally, we conclude the paper with some remarks in Section 7.

2. The Swift-Hohenberg equation and mixed weak form

This section aims to introduce the considered SH equation and give its mixed weak form.

2.1. The Swift-Hohenberg equation

The SH equation can be derived from minimizing Lyapunov functional that describes the configurational cost of periodic phase in contact with isotropic phase [5,23]

$$E(\phi)=\int_\Omega\left(\frac{1}{4}\phi^4+\frac{1-\epsilon}{2}\phi^2-|\nabla\phi|^2+\frac{1}{2}(\Delta\phi)^2\right)\mathrm{d}\mathbf{x}, \tag{2.1}$$

where $\phi=\phi(\mathbf{x},t)$ is the density field and ϵ is a positive constant, $\Omega=(0,L)\times(0,L)$ with $L>0$, and ∇ and Δ are the gradient and Laplacian operators, respectively.

The SH model is the non-conserved type dynamics as below

$$\begin{cases}\partial_t\phi=-M(\phi)\mu & \text{in } \Omega\times(0,T),\\ \phi(\mathbf{x},0)=\phi_0(\mathbf{x}) & \text{in } \Omega\end{cases} \tag{2.2}$$

with the periodic boundary condition. $M(\phi)$ is a mobility and μ is the chemical potential as

$$\mu=\frac{\delta E}{\delta\phi}=\Delta^2\phi-2\Delta\phi-\phi^3-(1-\epsilon)\phi,$$

where $\delta E/\delta\phi$ denotes the variational derivative of E with respect to ϕ.

For the aim of the mixed finite element discretization, we introduce a new function $w=-\Delta\phi$. Then the mixed form of (2.2) can be defined as

$$\begin{cases}\partial_t\phi-\Delta w=-2\Delta\phi-\phi^3-(1-\epsilon)\phi & \text{in } \Omega\times(0,T),\\ -\Delta\phi-w=0 & \text{in } \Omega\times(0,T),\\ \phi(\mathbf{x},0)=\phi_0(\mathbf{x}) & \text{in } \Omega\end{cases} \tag{2.3}$$

with $M(\phi)=1$.

2.2. Mixed weak form

Now, we present a mixed weakform for Eq. (2.3). We adopt the standard notation for Sobolev spaces $W^{s,p}(\Omega)$ along with their associated norms and seminorms (see, e.g., [1,3]). Specifically, we denote $H^s(\Omega) = W^{s,2}(\Omega)$ and define the space of functions with periodic boundary conditions as

$$H^s_{\text{per}}(\Omega) = \big\{v \in H^s(\Omega) \mid v \text{ satisfies periodic boundary conditions}\big\}.$$

The norms $\|v\|_s = \|v\|_{s,2,\Omega}$ and $\|v\|_0 = \|v\|_{0,2,\Omega}$ are given by

$$\|v\|_s^2 = \sum_{|\alpha|\le s} \|D^\alpha v\|_0^2, \quad \|v\|_0^2 = \int_\Omega |v|^2 \mathrm{d}\mathbf{x}.$$

Let X be a Banach space equipped with a norm $\|\cdot\|_X$ and a seminorm $|\cdot|_X$. We then introduce the Bochner space norms as follows:

$$C(0,T;X) = \left\{v : [0,T] \to X,\ \|v\|_{C(0,T;X)} = \sup_{t\in[0,T]} \|v(t)\|_X < \infty\right\},$$
$$L^2(0,T;X) = \left\{v : (0,T) \to X,\ \|v\|_{L^2(0,T;X)} = \left(\int_0^T \|v(t)\|_X^2 dt\right)^{1/2} < \infty\right\},$$
$$H^m(0,T;X) = \left\{v \in L^2(0,T;X) : \frac{\partial^j v}{\partial t^j} \in L^2(0,T;X),\ 1 \le j \le m\right\},$$

where the partial derivatives $\partial^j v/\partial t^j$ are defined in the sense of distributions on $(0,T)$. The norm in space $H^m(0,T;X)$ is defined by

$$\|v\|^2_{H^m(0,T;X)} = \int_0^T \sum_{j=0}^m \left\|\frac{\partial^j v}{\partial t^j}\right\|_X^2 \mathrm{d}t.$$

A weak solution to (2.3) is a pair (ϕ, w) solving the following problem.

Problem 2.1. Determine a pair $(\phi, w) \in L^\infty(0,T;H^1_{\text{per}}(\Omega)) \times L^2(0,T;H^1_{\text{per}}(\Omega))$, $\partial_t\phi \in L^2(0,T;H^{-1}_{\text{per}}(\Omega))$ such that for any $(\psi, v) \in H^1_{\text{per}}(\Omega) \times H^1_{\text{per}}(\Omega)$,

$$\begin{cases} (\partial_t\phi,\psi) + (\nabla w, \nabla\psi) = 2(\nabla\phi,\nabla\psi) - (\phi^3,\psi) - (1-\epsilon)(\phi,\psi) & \text{for} \quad t \in (0,T), \\ (\nabla\phi,\nabla v) - (w,v) = 0 & \text{for} \quad t \in (0,T), \\ (\phi(\mathbf{x},0),\psi) = (\phi_0(\mathbf{x}),\psi), \\ (\nabla\phi(\mathbf{x},0),\nabla v) - (w(\mathbf{x},0),v) = 0. \end{cases} \tag{2.4}$$

Similar to the method in [13, Theorem 3.2], we can give the following stability result.

Lemma 2.1. *Let $\phi_0, \theta_0 \in H^1_{\text{per}}(\Omega)$ and (ϕ, w), (θ, u) be the weak solutions of* (2.4) *with $\phi(\mathbf{x}, 0) = \phi_0$ and $\theta(\mathbf{x}, 0) = \theta_0$ a.e. Ω. Then*

$$\|\phi - \theta\|_{L^\infty(0,T;L^2(\Omega))} + \|w - u\|_{L^2(0,T;L^2(\Omega))} \leq C \|\phi_0 - \theta_0\|_0,$$

where $C = C(T) > 0$ is a constant.

Similarly to [13, Theorem 3.3], the regularity result is presented in the following lemma.

Lemma 2.2. *Let $\phi_0(\mathbf{x}) \in H^r_{\text{per}}(\Omega)$ and $r \geq 2$ be some integer. Then the weak solution $\phi : \Omega \times [0, T] \to \mathbb{R}$ and $w : \Omega \times [0, T] \to \mathbb{R}$ of the initial boundary value problem* (2.3) *has the regularity $\phi \in L^\infty(0, T; H^r(\Omega)) \cap L^2(0, T; H^{r+2}(\Omega))$, $\partial_t \phi \in L^2(0, T; H^{r-2}(\Omega))$ and $w \in L^2(0, T; H^r(\Omega))$.*

For the mixed form of the SH equation, the energy functional corresponding to (2.1) is defined as

$$E(\phi, w) = \int_\Omega \left\{ \frac{1}{4}\phi^4 + \frac{1-\epsilon}{2}\phi^2 - |\nabla\phi|^2 + \frac{1}{2}w^2 \right\} \mathrm{d}\mathbf{x}. \tag{2.5}$$

Similarly, we can derive the energy identity for (2.4).

Lemma 2.3. *Let $(\phi(\mathbf{x}, t), w(\mathbf{x}, t))$ be a weak solution of the mixed SH equation* (2.4)*, then we have the following energy identity:*

$$\frac{\mathrm{d}}{\mathrm{d}t} E(\phi, w) + \|\partial_t \phi\|_0^2 = 0 \quad \textit{a.e.} \quad t \in (0, T). \tag{2.6}$$

Proof. According to the second equation of (2.4), we have

$$\left(\frac{\partial w}{\partial t}, w \right) = \left(\nabla \frac{\partial \phi}{\partial t}, \nabla w \right). \tag{2.7}$$

Choosing $\psi = \partial_t \phi$ in the first equation of (2.4) and using (2.7) and the fact that

$$\left(\frac{\partial \phi}{\partial t}, \phi \right) = \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} \|\phi\|_0^2,$$

we get

$$\begin{aligned}
\left(\frac{\partial \phi}{\partial t}, \frac{\partial \phi}{\partial t} \right) &= -\left(\phi^3 + (1-\epsilon)\phi, \frac{\partial \phi}{\partial t} \right) + 2\left(\nabla\phi, \nabla \frac{\partial \phi}{\partial t} \right) - \left(\nabla w, \nabla \frac{\partial \phi}{\partial t} \right) \\
&= -\left(\phi^3 + (1-\epsilon)\phi, \frac{\partial \phi}{\partial t} \right) + 2\left(\nabla\phi, \frac{\partial (\nabla\phi)}{\partial t} \right) - \left(\frac{\partial w}{\partial t}, w \right) \\
&= -\frac{\mathrm{d}}{\mathrm{d}t} \int_\Omega \left(\frac{1}{4}\phi^4 + \frac{1-\epsilon}{2}\phi^2 - |\nabla\phi|^2 + \frac{1}{2}w^2 \right) \mathrm{d}\mathbf{x} \\
&= -\frac{\mathrm{d}}{\mathrm{d}t} E(\phi, w).
\end{aligned}$$

That is the desired result (2.6). □

3. Mixed finite element discretization in space

We first introduce a mixed finite element method in space for the Swift-Hohenberg equation (2.2) and derive error estimates for the semidiscrete form in this section.

3.1. Mixed finite element discretization

To perform the finite element discretization, we define a face-to-face partition $\mathcal{T}_h$ of the computational domain Ω into elements K, which can be triangles or rectangles such that

$$\overline{\Omega} = \bigcup_{K\in\mathcal{T}_h} K,$$

here, $h := \max_{K\in\mathcal{T}_h} h_K$ represents the global mesh size, where $h_K = \operatorname{diam} K$ is the diameter of element K. The radius of the inscribed circle of element K is denoted by τ_K. A set of partitions $\mathcal{T}_h$ is considered quasi-uniform if it satisfies certain regularity conditions (see, e.g., [3])

$$\begin{aligned}&\exists\sigma>0 \quad \text{such that} \quad h_K/\tau_K \le \sigma, \quad \forall K\in\mathcal{T}_h,\\ &\exists\beta>0 \quad \text{such that} \quad \max\{h/h_K, K\in\mathcal{T}_h\}\le\beta.\end{aligned}$$

Based on the partition $\mathcal{T}_h$, we build the finite element space V_h of piecewise polynomial functions

$$V_h := \Big\{v_h \in H^1_{\text{per}}(\Omega),\, v_h|_K \in \mathcal{P}_m(K) \ \text{ or } \ \mathcal{Q}_m(K),\ \forall K\in\mathcal{T}_h\Big\},$$

where $\mathcal{P}_m(K)$ denotes the space of polynomials with degree not greater than m and $\mathcal{Q}_m(K)$ denotes the space of polynomials with degree not greater than m in each variable.

Below we define the corresponding semidiscrete weak solution to (2.4).

Problem 3.1. Find $(\phi_h, w_h)\in L^\infty(0,T;V_h)\times L^2(0,T;V_h)$, $\partial_t\phi_h\in L^2(0,T;V_h)$ such that for any $(\psi_h, v_h)\in V_h\times V_h$,

$$\begin{cases} (\partial_t\phi_h,\psi_h)+(\nabla w_h,\nabla\psi_h) & \\ =2(\nabla\phi_h,\nabla\psi_h)-(\phi_h^3,\psi_h)-(1-\epsilon)(\phi_h,\psi_h) & \text{for} \quad t\in(0,T),\\ (\nabla\phi_h,\nabla v_h)-(w_h,v_h)=0 & \text{for} \quad t\in(0,T),\\ (\phi_h(\mathbf{x},0),\psi_h)=(\phi_0(\mathbf{x}),\psi_h), & \\ (\nabla\phi_h(\mathbf{x},0),\nabla v_h)-(w_h(\mathbf{x},0),v_h)=0. & \end{cases} \tag{3.1}$$

Theorem 3.1. *The semidiscrete scheme* (3.1) *has a unique solution.*

Proof. Let $N=\dim V_h$ and $\{\lambda_j,\psi_j\}_{1\le j\le N}$ be the eigenpair system of the following eigenvalue problem: Find $(\lambda,\psi)\in\mathbb{R}\times V_h$ such that $\|\psi\|_0=1$ and

$$(\nabla\psi,\nabla v)=\lambda(\psi,v),\quad \forall v\in V_h.$$

Then we have

$$(\psi_i, \psi_j) = \delta_{ij},$$

where δ_{ij} is the Kronecker delta function. Let

$$\phi_h(\mathbf{x}, t) = \sum_{j=1}^{N} a_j(t)\psi_j(\mathbf{x}), \quad w_h(\mathbf{x}, t) = \sum_{j=1}^{N} b_j(t)\psi_j(\mathbf{x}).$$

From the second equation in (3.1), we have $b_j(t) = \lambda_j a_j(t)$ for $j = 1, \ldots, N$. Then the first equation in (3.1) can be written in the following form:

$$\frac{\partial a_j(t)}{\partial t} + \lambda_j^2 a_j(t) = f_j\big(a_1(t), \ldots, a_N(t)\big), \quad j = 1, \ldots, N, \tag{3.2}$$

where all $f_j : \mathbb{R}^N \mapsto \mathbb{R}$ $(1 \le j \le N)$ are smooth and locally Lipschitz. Set

$$a_j(0) = \big(\phi_0(\mathbf{x}), \psi_j(\mathbf{x})\big), \quad j = 1, \ldots, N. \tag{3.3}$$

From the theory of ordinary differential equations, we know that there exists a unique smooth solution $(a_1(t), \ldots, a_N(t)), t \in [0, T]$ for (3.2) with initial conditions (3.3). □

For the semidiscrete weak solution (ϕ_h, w_h), we have the similar energy identities.

Lemma 3.1. *Let $(\phi_h(\mathbf{x}, t), w_h(\mathbf{x}, t))$ be a weak solution of* (3.1)*, then the following energy identity holds:*

$$\frac{\mathrm{d}}{\mathrm{d}t} E(\phi_h, w_h) + \|\partial_t \phi_h\|_0^2 = 0 \quad \textit{a.e.} \quad t \in (0, T). \tag{3.4}$$

Proof. In order to prove (3.4), we use the second equation of (3.1) to obtain

$$\left(\frac{\partial w_h}{\partial t}, w_h\right) = \left(\nabla \frac{\partial \phi_h}{\partial t}, \nabla w_h\right). \tag{3.5}$$

Using (3.5) and choosing $\psi_h = \partial_t \phi_h$ in the first equation of (3.1), we get

$$\begin{aligned}
\left(\frac{\partial \phi_h}{\partial t}, \frac{\partial \phi_h}{\partial t}\right) &= -\left(\phi_h^3 + (1-\epsilon)\phi_h, \frac{\partial \phi_h}{\partial t}\right) + 2\left(\nabla \phi_h, \nabla \frac{\partial \phi_h}{\partial t}\right) - \left(\nabla w_h, \nabla \frac{\partial \phi_h}{\partial t}\right) \\
&= -\left(\phi_h^3 + (1-\epsilon)\phi_h, \frac{\partial \phi_h}{\partial t}\right) + 2\left(\nabla \phi_h, \frac{\partial (\nabla \phi_h)}{\partial t}\right) - \left(\frac{\partial w_h}{\partial t}, w_h\right) \\
&= -\frac{\mathrm{d}}{\mathrm{d}t} \int_\Omega \left(\frac{1}{4}\phi_h^4 + \frac{1-\epsilon}{2}\phi_h^2 - |\nabla \phi_h|^2 + \frac{1}{2} w_h^2\right) \mathrm{d}\mathbf{x} \\
&= -\frac{\mathrm{d}}{\mathrm{d}t} E(\phi_h, w_h),
\end{aligned}$$

which is the desired result (3.4). □

3.2. Error estimates of semidiscrete form

In this subsection, we analyze the error estimate of the semidiscrete form (3.1). Based on the finite element space V_h, define the Ritz-projection P_h and the L^2-projection operator π_h by

$$(\nabla P_h u, \nabla v_h) = (\nabla u, \nabla v_h), \quad \forall v_h \in V_h, \tag{3.6}$$

and

$$(\pi_h u, v_h) = (u, v_h), \quad \forall v_h \in V_h, \tag{3.7}$$

respectively. Let (ϕ_h, w_h) be the solution of (3.1) and define

$$\xi_\phi := \phi_h - P_h\phi, \quad \eta_\phi := P_h\phi - \phi, \quad \xi_w := w_h - P_h w, \quad \eta_w := P_h w - w, \tag{3.8}$$

$$\tilde{\xi}_\phi := \phi_h - \pi_h\phi, \quad \tilde{\eta}_\phi := \pi_h\phi - \phi, \quad \tilde{\xi}_w := w_h - \pi_h w, \quad \tilde{\eta}_w := \pi_h w - w. \tag{3.9}$$

Then we have

$$\phi_h - \phi = \xi_\phi + \eta_\phi, \quad w_h - w = \xi_w + \eta_w, \tag{3.10}$$

$$\phi_h - \phi = \tilde{\xi}_\phi + \tilde{\eta}_\phi, \quad w_h - w = \tilde{\xi}_w + \tilde{\eta}_w. \tag{3.11}$$

It follows from (3.1), (3.6) and (3.7) that

$$\begin{aligned} &(\nabla(\phi_h - P_h\phi), \nabla v_h) \\ &= (\nabla\phi_h, \nabla v_h) - (\nabla\phi, \nabla v_h) \\ &= (w_h - w, v_h) = (w_h - \pi_h w, v_h), \quad \forall v_h \in V_h, \end{aligned}$$

which leads to

$$\|\nabla\xi_\phi\|_0^2 \le \|\tilde{\xi}_w\|_0 \|\xi_\phi\|_0. \tag{3.12}$$

On the other hand, it follows from (2.4) and (3.1) that

$$\begin{aligned} (\nabla\xi_w, \nabla\xi_\phi) &= (\nabla\xi_w, \nabla(\phi_h - P_h\phi)) \\ &= (\nabla\xi_w, \nabla(\phi_h - \phi)) = (\xi_w, w_h - w) \\ &= (\xi_w, w_h - \pi_h w) = (\xi_w, w_h - P_h w) + (\xi_w, P_h w - \pi_h w) \\ &= \|\xi_w\|_0^2 + (\xi_w, \eta_w - \tilde{\eta}_w). \end{aligned} \tag{3.13}$$

In our analysis, we also need the following two formulas:

$$\begin{aligned} (\tilde{\xi}_\phi, \xi_\phi) &= (\phi_h - \pi_h\phi, \xi_\phi) = (\phi_h - \phi, \xi_\phi) = \|\xi_\phi\|_0^2 + (\eta_\phi, \xi_\phi) \\ &\le \frac{3}{2}\|\xi_\phi\|_0^2 + \frac{1}{2}\|\eta_\phi\|_0^2, \end{aligned} \tag{3.14}$$

$$(\tilde{\xi}_w, \xi_\phi) = (w_h - \pi_h w, \xi_\phi) = (w_h - w, \xi_\phi) = (\xi_w, \xi_\phi) + (\eta_w, \xi_\phi). \tag{3.15}$$

Lemma 3.2. *Suppose* (ϕ, w) *is the solution of* (2.4) *and* (ϕ_h, w_h) *is the solution of* (3.1)*, then we have the following error estimate:*

$$\begin{aligned}&\|\phi(\mathbf{x},T)-\phi_h(\mathbf{x},T)\|_0^2+\int_0^T\|w-w_h\|_0^2\mathrm{d}t\\&\leq C\left(\int_0^T\left(\|\partial_t\eta_\phi\|_0^2+\|\eta_\phi\|_0^2+\|\eta_w\|_0^2+\|\tilde{\eta}_w\|_0^2\right)\mathrm{d}t+\|\phi_0(\mathbf{x})-\phi_h(\mathbf{x},0)\|_0^2\right).\end{aligned}\tag{3.16}$$

Proof. From Eqs. (2.4), (3.1), (3.8), and (3.9), along with the regularity result in Lemma 2.2 ($\phi \in L^\infty(0,T;W^{1,\infty}(\Omega))$) and the application of the Cauchy-Schwarz inequality, it follows that

$$\begin{aligned}&(\partial_t\xi_\phi,\xi_\phi)+(1-\epsilon)(\tilde{\xi}_\phi,\xi_\phi)+(\nabla\xi_w,\nabla\xi_\phi)+2(\tilde{\xi}_w,\xi_\phi)+\left(|\phi_h|^2\phi_h-|P_h\phi|^2P_h\phi,\xi_\phi\right)\\&=-(\partial_t\eta_\phi,\xi_\phi)+\left(|\phi|^2\phi-|P_h\phi|^2P_h\phi,\xi_\phi\right)\\&\leq|(\partial_t\eta_\phi,\xi_\phi)|+|(|\phi|^2(\phi-P_h\phi),\xi_\phi)|+|((|\phi|^2-|P_h\phi|^2)P_h\phi,\xi_\phi)|\\&\leq\|\partial_t\eta_\phi\|_0\|\xi_\phi\|_0+C\|\eta_\phi\|_0\|\xi_\phi\|_0.\end{aligned}\tag{3.17}$$

For the nonlinear term, we obtain

$$\begin{aligned}&\left(|\phi_h|^2\phi_h-|P_h\phi|^2P_h\phi,\xi_\phi\right)\\&=\left(|\phi_h|^2\phi_h-|P_h\phi|^2P_h\phi,\phi_h-P_h\phi\right)\\&=\|\phi_h\|_{0,4}^4+\|P_h\phi\|_{0,4}^4-\left(|\phi_h|^2\phi_h,P_h\phi\right)-\left(|P_h\phi|^2P_h\phi,\phi_h\right)\\&\geq\frac{1}{2}\|\phi_h\|_{0,4}^4+\frac{1}{2}\|P_h\phi\|_{0,4}^4-\left(|\phi_h|^2,|P_h\phi|^2\right)\geq0.\end{aligned}\tag{3.18}$$

By combining (3.17) and (3.18), we obtain the following estimates:

$$\begin{aligned}&(\partial_t\xi_\phi,\xi_\phi)+(1-\epsilon)(\tilde{\xi}_\phi,\xi_\phi)+(\nabla\xi_w,\nabla\xi_\phi)+2(\tilde{\xi}_w,\xi_\phi)\\&\leq\|\partial_t\eta_\phi\|_0\|\xi_\phi\|_0+C\|\eta_\phi\|_0\|\xi_\phi\|_0.\end{aligned}\tag{3.19}$$

From (3.12)-(3.14), and (3.19), we can deduce that

$$\begin{aligned}&(\partial_t\xi_\phi,\xi_\phi)+\|\xi_w\|_0^2\\&\leq\|\partial_t\eta_\phi\|\|\xi_\phi\|+C\|\eta_\phi\|\|\xi_\phi\|+(1-\epsilon)|(\tilde{\xi}_\phi,\xi_\phi)|+2|(\tilde{\xi}_w,\xi_\phi)|+|(\xi_w,\eta_w-\tilde{\eta}_w)|\\&\leq\frac{1}{2}\|\partial_t\eta_\phi\|_0^2+\frac{1}{2}\|\xi_\phi\|_0^2+C\|\eta_\phi\|_0^2+C\|\xi_\phi\|_0^2+(1-\epsilon)\left(\frac{3}{2}\|\xi_\phi\|_0^2+\frac{1}{2}\|\eta_\phi\|_0^2\right)\\&\quad+2\left(\frac{1}{8}\|\xi_w\|_0^2+\frac{1}{2}\|\eta_w\|_0^2+\frac{1}{2}\|\tilde{\eta}_w\|_0^2+3\|\xi_\phi\|_0^2\right)+\frac{1}{4}\|\xi_w\|_0^2+2\|\eta_w\|_0^2+2\|\tilde{\eta}_w\|_0^2,\end{aligned}$$

which implies that

$$\frac{1}{2}\frac{\mathrm{d}}{\mathrm{d}t}\|\xi_\phi\|_0^2+\frac{1}{2}\|\xi_w\|_0^2\leq C\left(\|\partial_t\eta_\phi\|_0^2+\|\eta_\phi\|_0^2+\|\eta_w\|_0^2+\|\tilde{\eta}_w\|_0^2\right)+C\|\xi_\phi\|_0^2.$$

Using Gronwall's inequality gives

$$\begin{aligned}&\|\xi_\phi(\mathbf{x},T)\|_0^2+\int_0^T\|\xi_w\|_0^2\mathrm{d}t\\&\leq C\left(\int_0^T\left(\|\partial_t\eta_\phi\|_0^2+\|\eta_\phi\|_0^2+\|\eta_w\|_0^2+\|\tilde{\eta}_w\|_0^2\right)\mathrm{d}t+\|\xi_\phi(\mathbf{x},0)\|_0^2\right).\end{aligned}$$

By applying the triangle inequality and utilizing (3.10) through (3.11), we can achieve the desired result (3.16). □

Now we are ready to give the error estimate of the semi-discrete scheme (2.4).

Theorem 3.2. *Suppose* (ϕ,w) *is the solution of* (2.4) *and* $\phi_0\in H_{per}^{m+1}(\Omega)$. *Then the finite element approximation* (ϕ_h,w_h) *of* (3.1) *has the following error estimate:*

$$\begin{aligned}&\|\phi(\mathbf{x},T)-\phi_h(\mathbf{x},T)\|_0^2+\int_0^T\|w-w_h\|_0^2\mathrm{d}t\\&\leq Ch^{2m}\left(\int_0^T\left(\|\partial_t\phi\|_m^2+\|\phi\|_m^2+\|w\|_m^2\right)\mathrm{d}t+\|\phi_0\|_m^2\right).\end{aligned}$$

Proof. We take $\phi_h(\mathbf{x},0)=P_h\phi_0$. Combining Lemmas 2.2 and 3.2 and the error estimates

$$\begin{aligned}&\|\partial_t\eta_\phi\|_0\leq Ch^m\|\partial_t\phi\|_m, && \|\eta_\phi\|_0\leq Ch^m\|\phi\|_m,\\&\|\eta_w\|_0\leq Ch^m\|w\|_m, && \|\tilde{\eta}_w\|_0\leq Ch^m\|w\|_m,\\&\|\phi_0-\phi_h(\mathbf{x},0)\|_0\leq Ch^m\|\phi_0\|_m,\end{aligned}$$

we have

$$\begin{aligned}&\|\phi(\mathbf{x},T)-\phi_h(\mathbf{x},T)\|_0^2+\int_0^T\|w-w_h\|_0^2\mathrm{d}t\\&\leq Ch^{2m}\left(\int_0^T\left(\|\partial_t\phi\|_m^2+\|\phi\|_m^2+\|w\|_m^2\right)\mathrm{d}t+\|\phi_0\|_m^2\right).\end{aligned}$$

The proof is complete. □

Remark 3.1. Unlike typical error estimates for parabolic equations, which usually provide a convergence order of $(m+1)$, the convergence rate derived above is only of m-th order. This reduction in the convergence rate is attributed to the challenges posed by the second-order nonlinear term. Addressing this discrepancy will require further investigation.

4. Fully discrete scheme and unconditional energy stability

In this section, we employ the second-order Crank-Nicolson scheme for time discretization. Utilizing the CN approximation results in the following fully discrete scheme for (2.4).

Problem 4.1. Given $(\phi_h^n, w_h^n) \in V_h \times V_h$, find $(\phi_h^{n+1}, w_h^{n+1}) \in V_h \times V_h$ such that for any $(\psi_h, v_h) \in V_h \times V_h$,

$$\begin{cases} \big(\mathrm{D}_\tau \phi_h^{n+1}, \psi_h\big) = -\big(\nabla w_h^{n+1/2}, \nabla \psi_h\big) + 2\big(\nabla \phi_h^{n+1/2}, \nabla \psi_h\big) \\ \qquad\qquad - \big(\zeta_h^{n+1/2}, \psi_h\big) - (1-\epsilon)\big(\phi_h^{n+1/2}, \psi_h\big), \\ \big(w_h^{n+1}, v_h\big) = \big(\nabla \phi_h^{n+1}, \nabla v_h\big) \end{cases} \tag{4.1}$$

with

$$\mathrm{D}_\tau \phi_h^{n+1} = \frac{\phi_h^{n+1} - \phi_h^n}{\tau}, \quad w_h^{n+1/2} = \frac{w_h^{n+1} + w_h^n}{2},$$
$$\phi_h^{n+1/2} = \frac{\phi_h^{n+1} + \phi_h^n}{2}, \quad \zeta_h^{n+1/2} = \frac{|\phi_h^{n+1}|^2 + |\phi_h^n|^2}{2} \phi_h^{n+1/2}.$$

Lemma 4.1 ([2]). *Let α, B and a_i, b_i, c_i, γ_i, for integer $i \geq 0$, be nonnegative numbers such that*

$$a_n + \alpha \sum_{i=0}^{n} b_i \leq \alpha \sum_{i=0}^{n} \gamma_i a_i + \alpha \sum_{i=0}^{n} c_i + B \quad \textit{for} \quad n \geq 0.$$

Assume that $\alpha\gamma_i < 1$ for all $i \geq 0$, and set $\sigma_i = (1 - \alpha\gamma_i)^{-1}$. Then we have

$$a_n + \alpha \sum_{i=0}^{n} b_i \leq \exp\left\{\alpha \sum_{i=0}^{n} \gamma_i \sigma_i\right\} \left(\alpha \sum_{i=0}^{n} c_i + B\right) \quad \textit{for} \quad n \geq 0.$$

Lemma 4.2. *Suppose (ϕ_h^n, w_h^n) and (φ_h^n, u_h^n) are two solutions of* (4.1) *with initial values $\phi(\mathbf{x}, 0) = \phi_0$ and $\varphi(\mathbf{x}, 0) = \varphi_0$, respectively. Let $T = N\tau$, when τ is small enough, we have*

$$\begin{aligned} &\big\|\phi_h^N - \varphi_h^N\big\|_0^2 + \frac{\tau}{4} \sum_{n=1}^{N-1} \big\|(w_h^{n+1} + w_h^n) - (u_h^{n+1} + u_h^n)\big\|_0^2 \\ &\leq C\|\phi(\mathbf{x}, 0) - \varphi(\mathbf{x}, 0)\|_0^2. \end{aligned} \tag{4.2}$$

Proof. Set $f_h^n = \phi_h^n - \varphi_h^n$ and $g_h^n = w_h^n - u_h^n$. We have

$$\begin{cases} \quad \big(\mathrm{D}_\tau f_h^{n+1}, f_h^{n+1} + f_h^n\big) \\ = -\left(\nabla g_h^{n+1/2}, \nabla(f_h^{n+1} + f_h^n)\right) - \left(\nu_h^{n+1/2}, f_h^{n+1} + f_h^n\right) \\ \quad\ +2\left(\nabla f_h^{n+1/2}, \nabla(f_h^{n+1} + f_h^n)\right) - (1-\epsilon)\left(f_h^{n+1/2}, f_h^{n+1} + f_h^n\right), \\ \quad (g_h^{n+1}, g_h^{n+1}) = (\nabla f_h^{n+1}, \nabla g_h^{n+1}) \end{cases} \tag{4.3}$$

with

$$\mathrm{D}_\tau f_h^{n+1} = \frac{f_h^{n+1} - f_h^n}{\tau}, \quad g_h^{n+1/2} = \frac{g_h^{n+1} + g_h^n}{2}, \quad f_h^{n+1/2} = \frac{f_h^{n+1} + f_h^n}{2},$$
$$\nu_h^{n+1/2} = \frac{(|\phi_h^{n+1}|^2 + |\phi_h^n|^2)(\phi_h^{n+1} + \phi_h^n)}{4} - \frac{(|\varphi_h^{n+1}|^2 + |\varphi_h^n|^2)(\varphi_h^{n+1} + \varphi_h^n)}{4}.$$

According to the same argument in (3.18), we assume

$$\left(\nu_h^{n+1/2}, f_h^{n+1}+f_h^n\right) \geq 0.$$

Then the following estimates hold:

$$\begin{aligned}
0 \geq & \frac{\|f_h^{n+1}\|_0^2-\|f_h^n\|_0^2}{\tau}+\left(\frac{\nabla g_h^{n+1}+\nabla g_h^n}{2}, \nabla(f_h^{n+1}+f_h^n)\right) \\
& -2\left(\frac{\nabla f_h^{n+1}+\nabla f_h^n}{2}, \nabla(f_h^{n+1}+f_h^n)\right)+(1-\epsilon)\frac{\|f_h^{n+1}+f_h^n\|_0^2}{2} \\
= & \frac{\|f_h^{n+1}\|_0^2-\|f_h^n\|_0^2}{\tau}+\frac{1}{2}\left(g_h^{n+1}+g_h^n, g_h^{n+1}+g_h^n\right)+\frac{1-\epsilon}{2}\left\|f_h^{n+1}+f_h^n\right\|_0^2 \\
& -\left(g_h^{n+1}+g_h^n, f_h^{n+1}+f_h^n\right) \\
\geq & \frac{\|f_h^{n+1}\|_0^2-\|f_h^n\|_0^2}{\tau}+\frac{1}{2}\left\|g_h^{n+1}+g_h^n\right\|_0^2+\frac{1-\epsilon}{2}\left\|f_h^{n+1}+f_h^n\right\|_0^2 \\
& -\frac{1}{4}\left\|g_h^{n+1}+g_h^n\right\|_0^2-\left\|f_h^{n+1}+f_h^n\right\|_0^2 \\
\geq & \frac{\|f_h^{n+1}\|_0^2-\|f_h^n\|_0^2}{\tau}+\frac{1}{4}\left\|g_h^{n+1}+g_h^n\right\|_0^2-(1+\epsilon)(\|f_h^{n+1}\|_0^2+\|f_h^n\|_0^2),
\end{aligned}$$

that is

$$\left\|f_h^{n+1}\right\|_0^2-\left\|f_h^n\right\|_0^2+\frac{\tau}{4}\left\|g_h^{n+1}+g_h^n\right\|_0^2 \leq (1+\epsilon)\tau(\|f_h^{n+1}\|_0^2+\|f_h^n\|_0^2).$$

Then we obtain

$$\left\|f_h^N\right\|_0^2+\frac{\tau}{4}\sum_{n=0}^{N-1}\left\|g_h^{n+1}+g_h^n\right\|_0^2 \leq 2(1+\epsilon)\tau\sum_{n=0}^{N}\|f_h^n\|_0^2+\|f_h^0\|_0^2.$$

The desired result (4.2) can be derived by combining the aforementioned inequality with Lemma 4.1, under the condition that $\tau < 1/(2(1+\epsilon))$. □

To establish the existence of a solution for the fully discrete scheme (4.1), we invoke the Brouwer fixed-point theorem, which can be stated as follows.

Lemma 4.3 ([10]). *Let H be a finite-dimensional Hilbert space equipped with an inner product $(\cdot,\cdot)$ and the induced norm $\|\cdot\|_H$. Suppose L is a continuous mapping from H to itself, satisfying $(L(\zeta),\zeta) > 0$ for all $\zeta \in H$ with $\|\zeta\|_H = \beta > 0$. Then there exists an element $\zeta^* \in H$ such that $L(\zeta^*) = 0$ and $\|\zeta^*\|_H \leq \beta$.*

Theorem 4.1. *The fully discretized scheme* (4.1) *has a unique solution.*

Proof. We set

$$\zeta = \frac{\phi_h^{n+1}+\phi_h^n}{2}, \quad \rho = \frac{w_h^{n+1}+w_h^n}{2},$$

and define $L(\zeta)$ such that

$$\begin{aligned}\big(L(\zeta),\psi_h\big) &= (\zeta,\psi_h)-(\phi_h^n,\psi_h)+\tau(\nabla\rho,\nabla\psi_h)+\tau(1-\epsilon)(\zeta,\psi_h)\\ &\quad -2\tau(\nabla\zeta,\nabla\psi_h)+\tau\left(\frac{(|2\zeta-\phi_h^n|^2+|\phi_h^n|^2)\zeta}{2},\psi_h\right),\quad \forall\psi_h\in V_h.\end{aligned}$$

Since the Young and Hölder inequalities, the following inequalities hold:

$$\begin{aligned}\big(L(\zeta),\zeta\big) &= (\zeta,\zeta)-(\phi_h^n,\zeta)+\tau(\rho,\rho)+\tau(1-\epsilon)(\zeta,\zeta)-2\tau(\rho,\zeta)\\ &\quad +\tau\big(2|\zeta|^2-2\zeta\phi_h^n+|\phi_h^n|^2,|\zeta|^2\big)\\ &\geq \|\zeta\|_0^2-\frac{1}{2}\|\phi_h^n\|_0^2-\frac{1}{2}\|\zeta\|_0^2+\tau\|\rho\|_0^2+\tau(1-\epsilon)\|\zeta\|_0^2-\tau\|\rho\|_0^2-\tau\|\zeta\|_0^2\\ &\quad +2\tau\|\zeta\|_{0,4}^4-2\tau(\zeta^3,\phi_h^n)+\tau(|\phi_h^n|^2,|\zeta|^2)\\ &\geq \|\zeta\|_0^2-\frac{1}{2}\|\phi_h^n\|_0^2-\frac{1}{2}\|\zeta\|_0^2+\tau\|\rho\|_0^2+\tau(1-\epsilon)\|\zeta\|_0^2-\tau\|\rho\|_0^2-\tau\|\zeta\|_0^2\\ &\quad +2\tau\|\zeta\|_{0,4}^4-\frac{3\tau}{2}\|\zeta\|_{0,4}^4-\frac{\tau}{2}\|\phi_h^n\|_{0,4}^4+\tau(|\phi_h^n|^2,|\zeta|^2)\\ &\geq \left(\frac{1}{2}-\epsilon\tau\right)\|\zeta\|_0^2-\frac{1}{2}\big(\|\phi_h^n\|_0^2+2\tau\|\phi_h^n\|_{0,4}^4\big).\end{aligned}$$

Let $\alpha=(\|\phi_h^n\|_0^2+2\tau\|\phi_h^n\|_{0,4}^4)/(1-2\epsilon\tau)$. For any $\zeta\in V_h$ with $\|\zeta\|_0^2=\alpha$, it holds that $(L(\zeta),\zeta)\geq 0$. According to Lemma 4.3, there exists at least one solution ζ^* such that $\|\zeta^*\|_0^2\leq\alpha$. This implies that the existence of ϕ_h^{n+1} holds. The uniqueness of solution can be established by Lemma 4.2. □

Theorem 4.2 (Unconditional Energy Stability)**.** *The fully discrete form* (4.1) *is unconditionally energy-stable with respect to the initial energy. Specifically, the following inequality holds for any $\tau>0$:*

$$E(\phi_h^{n+1},w_h^{n+1})\leq E(\phi_h^n,w_h^n),\tag{4.4}$$

where E is the energy and defined in (2.5)*, and (ϕ_h^n,w_h^n) represents the solution of* (4.1) *at t_n. Moreover, the discrete form of the energy identity, analogous to* (3.4)*, is given by*

$$\frac{E(\phi_h^{n+1},w_h^{n+1})-E(\phi_h^n,w_h^n)}{\tau}+\left\|\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right\|_0^2=0.\tag{4.5}$$

Proof. Choosing $\psi_h=(\phi_h^{n+1}-\phi_h^n)/\tau$ in (4.1) and combining

$$(\nabla\phi_h^n,\nabla w_h^n)=(w_h^n,w_h^n)\quad\text{and}\quad(w_h^n,\phi_h^n)=(\nabla\phi_h^n,\nabla\phi_h^n),$$

we get

$$\left(\frac{\phi_h^{n+1}-\phi_h^n}{\tau},\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right)=-\left(\frac{\nabla w_h^{n+1}+\nabla w_h^n}{2},\frac{\nabla\phi_h^{n+1}-\nabla\phi_h^n}{\tau}\right)$$

$$
\begin{aligned}
&+2\left(\frac{\nabla\phi_h^{n+1}+\nabla\phi_h^n}{2},\frac{\nabla\phi_h^{n+1}-\nabla\phi_h^n}{\tau}\right)-(1-\epsilon)\left(\frac{\phi_h^{n+1}+\phi_h^n}{2},\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right)\\
&-\left(\frac{(|\phi_h^{n+1}|^2+|\phi_h^n|^2)(\phi_h^{n+1}+\phi_h^n)}{4},\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right)\\
=&-\left(\frac{w_h^{n+1}+w_h^n}{2},\frac{w_h^{n+1}-w_h^n}{\tau}\right)+2\left(\frac{\nabla\phi_h^{n+1}+\nabla\phi_h^n}{2},\frac{\nabla\phi_h^{n+1}-\nabla\phi_h^n}{\tau}\right)\\
&-(1-\epsilon)\left(\frac{\phi_h^{n+1}+\phi_h^n}{2},\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right)-\left(\frac{|\phi_h^{n+1}|^2+|\phi_h^n|^2}{4},\frac{|\phi_h^{n+1}|^2-|\phi_h^n|^2}{\tau}\right)\\
=&-\frac{1}{2\tau}\big(\|w_h^{n+1}\|_0^2-\|w_h^n\|_0^2\big)+\frac{1}{\tau}\big(\|\nabla\phi_h^{n+1}\|_0^2-\|\nabla\phi_h^n\|_0^2\big)\\
&-\frac{1-\epsilon}{2\tau}\big(\|\phi_h^{n+1}\|_0^2-\|\phi_h^n\|_0^2\big)-\frac{1}{4\tau}\big(\|\phi_h^{n+1}\|_{0,4}^4-\|\phi_h^n\|_{0,4}^4\big). \qquad (4.6)
\end{aligned}
$$

Similarly, from the energy definition (2.5) and (4.6), we have

$$
E\big(\phi_h^{n+1},w_h^{n+1}\big)-E\big(\phi_h^n,w_h^n\big)=-\tau\left\|\frac{\phi_h^{n+1}-\phi_h^n}{\tau}\right\|_0^2.
$$

Thus, we obtain the desired energy stable result (4.4) and the discrete energy identity (4.5). □

5. Error estimates for the fully discrete scheme

In this section, we present the error estimates for the fully discrete scheme (4.1).

5.1. Main results

Here, we state the main results concerning the error estimates for the fully discrete solution of (4.1), which will be proven in the subsequent subsection.

Theorem 5.1. *Suppose* (ϕ,w) *is the solution of* (2.4)*,* $\phi_0\in H^m_{\rm per}(\Omega)$ *and the conditions in Lemma* 5.1 *are satisfied. If the time step* τ *is sufficiently small, then the finite element approximation* (ϕ_h^n,w_h^n) *of* (4.1) *satisfies the following error estimate:*

$$
\begin{aligned}
&\big\|\phi(\mathbf{x},T)-\phi_h^N\big\|_0^2+\frac{1}{4}\sum_{n=0}^{N-1}\tau\big\|w(\mathbf{x},t_{n+1})+w(\mathbf{x},t_n)-(w_h^{n+1}+w_h^n)\big\|_0^2\\
&\le C_{\epsilon,\phi,w}(h^{2m}+\tau^4), \qquad (5.1)
\end{aligned}
$$

where the constant $C_{\epsilon,\phi,w}$ *is independent of the mesh size* h *and the time step* τ*, but depends on* ϵ*,* ϕ*,* ϕ_0*, and* w*.*

5.2. The proof of Theorem 5.1

For simplicity, we set

$$\phi^n := \phi(\mathbf{x}, t_n), \quad \phi^{n+1/2} := \phi\left(\mathbf{x}, \frac{t_n + t_{n+1}}{2}\right),$$
$$w^n := w(\mathbf{x}, t_n), \quad w^{n+1/2} := w\left(\mathbf{x}, \frac{t_n + t_{n+1}}{2}\right).$$

To give the error estimate, we now introduce some useful notation. Let

$$\xi_\phi^n := \phi_h^n - P_h\phi^n, \quad \eta_\phi^n := P_h\phi^n - \phi^n, \tag{5.2}$$
$$\xi_w^n := w_h^n - P_h w^n, \quad \eta_w^n := P_h w^n - w^n, \tag{5.3}$$
$$\tilde{\xi}_\phi^n := \phi_h^n - \pi_h\phi^n, \quad \tilde{\eta}_\phi^n := \pi_h\phi^n - \phi^n, \tag{5.4}$$
$$\tilde{\xi}_w^n := w_h^n - \pi_h w^n, \quad \tilde{\eta}_w^n := \pi_h w^n - w^n. \tag{5.5}$$

Lemma 5.1. *Suppose (ϕ, w) is the solution of (2.4), $\phi_0 \in H^6_{\mathrm{per}}(\Omega)$, $\phi_h^0 = P_h\phi_0$ and $T = N\tau$. If the time step τ is sufficiently small, then the finite element approximation (ϕ_h^n, w_h^n) of (4.1) satisfies the following error estimate, using the notation defined in (5.2)-(5.5):*

$$\begin{aligned}
&\left\|\xi_\phi^N\right\|_0^2 + \frac{1}{4}\sum_{n=0}^{N-1}\tau\left\|\xi_w^{n+1} + \xi_w^n\right\|_0^2 \\
&\le C\tau^4\int_0^T \|\phi_{ttt}\|_0^2 \mathrm{d}s + C(1-\epsilon)^2\tau^4\int_0^T \|\phi_{ttt}\|_0^2 \mathrm{d}s + C\tau^4\int_0^T \|\phi_{tt}\|_0^2 \mathrm{d}s \\
&\quad + C\tau^4\int_0^T \|\nabla w_{tt}\|_0^2 \mathrm{d}s + C\int_0^T \|\partial_t\eta_\phi\|_0^2 \mathrm{d}s + C\sum_{n=0}^{N-1}\tau\left\|\eta_w^{n+1} + \eta_w^n\right\|_0^2 \\
&\quad + C\sum_{n=0}^{N-1}\tau\left\|\tilde{\eta}_w^{n+1} + \tilde{\eta}_w^n\right\|_0^2 + C\sum_{n=0}^{N-1}\tau\left\|\eta_\phi^{n+1} + \eta_\phi^n\right\|_0^2 + \left\|\xi_\phi^0\right\|_0^2,
\end{aligned} \tag{5.6}$$

where the constant C is independent of the mesh size h and the time step τ but dependent on ϕ.

Proof. Using the regularity results [13, Theorem 3.3], we get $\phi_{ttt} \in L^2(0,T;L^2(\Omega))$ and $\phi \in L^2(0,T;W^{1,\infty}(\Omega))$ when $\phi_0 \in H^6_{per}(\Omega)$. Noting that $(\phi(\mathbf{x},t), w(\mathbf{x},t))$ and $(\phi_h^{n+1}, w_h^{n+1})$ are the solutions of (2.4) and (4.1) respectively, we have

$$\begin{aligned}
&\left(\frac{\xi_\phi^{n+1} - \xi_\phi^n}{\tau}, \xi_\phi^{n+1} + \xi_\phi^n\right) + (1-\epsilon)\left(\frac{\tilde{\xi}_\phi^{n+1} + \tilde{\xi}_\phi^n}{2}, \xi_\phi^{n+1} + \xi_\phi^n\right) \\
&\quad + \left(\frac{\nabla\xi_w^{n+1} + \nabla\xi_w^n}{2}, \nabla\xi_\phi^{n+1} + \nabla\xi_\phi^n\right) - 2\left(\frac{\nabla\xi_\phi^{n+1} + \nabla\xi_\phi^n}{2}, \nabla\xi_\phi^{n+1} + \nabla\xi_\phi^n\right)
\end{aligned}$$

$$
\begin{aligned}
&+\left(\frac{(|\phi_h^{n+1}|^2+|\phi_h^n|^2)(\phi_h^{n+1}+\phi_h^n)}{4},\xi_\phi^{n+1}+\xi_\phi^n\right)\\
&-\left(\frac{(|P_h\phi^{n+1}|^2+|P_h\phi^n|^2)(P_h\phi^{n+1}+P_h\phi^n)}{4},\xi_\phi^{n+1}+\xi_\phi^n\right)\\
&=I_1+I_2+I_3+I_4+I_5,
\end{aligned}\tag{5.7}
$$

where

$$
\begin{aligned}
I_1&=\left(\partial_t\phi^{n+1/2}-\frac{P_h\phi^{n+1}-P_h\phi^n}{\tau},\xi_\phi^{n+1}+\xi_\phi^n\right),\\
I_2&=(1-\epsilon)\left(\phi^{n+1/2}-\frac{\pi_h\phi^{n+1}+\pi_h\phi^n}{2},\xi_\phi^{n+1}+\xi_\phi^n\right),\\
I_3&=\left(\nabla w^{n+1/2}-\frac{\nabla P_h w^{n+1}+\nabla P_h w^n}{2},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right),\\
I_4&=2\left(\nabla\phi^{n+1/2}-\frac{\nabla P_h w^{n+1}+\nabla P_h w^n}{2},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right),\\
I_5&=\left(|\phi^{n+1/2}|^2\phi^{n+1/2},\xi_\phi^{n+1}+\xi_\phi^n\right)\\
&\quad-\left(\frac{(|P_h\phi^{n+1}|^2+|P_h\phi^n|^2)(P_h\phi^{n+1}+P_h\phi^n)}{4},\xi_\phi^{n+1}+\xi_\phi^n\right).
\end{aligned}
$$

Then we need estimate terms $I_1,I_2,\cdots,I_5$. For I_1, using Taylor expansion, (3.8) and Cauchy-Schwarz inequality, we have

$$
\begin{aligned}
|I_1|&\le\left|\left(\partial_t\phi^{n+1/2}-\frac{\phi^{n+1}-\phi^n}{\tau},\xi_\phi^{n+1}+\xi_\phi^n\right)\right|\\
&\quad+\left|\left(\frac{(\phi^{n+1}-P_h\phi^{n+1})-(\phi^n-P_h\phi^n)}{\tau},\xi_\phi^{n+1}+\xi_\phi^n\right)\right|\\
&=\left|\left(\partial_t\phi^{n+1/2}-\frac{\phi^{n+1}-\phi^n}{\tau},\xi_\phi^{n+1}+\xi_\phi^n\right)\right|+\frac{1}{\tau}\left|\left(\int_{t_n}^{t_{n+1}}\partial_t\eta_\phi\mathrm{d}s,\xi_\phi^{n+1}+\xi_\phi^n\right)\right|\\
&\le C\tau\left(\int_{t_n}^{t_{n+1}}\|\phi_{ttt}\|_0\mathrm{d}s\right)\left\|\xi_\phi^{n+1}+\xi_\phi^n\right\|_0+\frac{1}{\tau}\left(\int_{t_n}^{t_{n+1}}\|\partial_t\eta_\phi\|_0\mathrm{d}s\right)\left\|\xi_\phi^{n+1}+\xi_\phi^n\right\|_0\\
&\le C\tau^3\int_{t_n}^{t_{n+1}}\|\phi_{ttt}\|_0^2\mathrm{d}s+\frac{C}{\tau}\int_{t_n}^{t_{n+1}}\|\partial_t\eta_\phi\|_0^2\mathrm{d}s+\|\xi_\phi^{n+1}\|_0^2+\|\xi_\phi^n\|_0^2,
\end{aligned}\tag{5.8}
$$

where C is a constant and independent of ϵ, h and τ. From (3.7), the Cauchy-Schwarz inequality and triangle inequality, the following estimate holds:

$$
\begin{aligned}
|I_2|&=\left|(1-\epsilon)\left(\phi^{n+1/2}-\frac{\pi_h\phi^{n+1}+\pi_h\phi^n}{2},\xi_\phi^{n+1}+\xi_\phi^n\right)\right|\\
&\le C(1-\epsilon)\tau\int_{t_n}^{t_{n+1}}\|\phi_{tt}\|_0\mathrm{d}s\left\|\xi_\phi^{n+1}+\xi_\phi^n\right\|_0
\end{aligned}
$$

$$\leq C(1-\epsilon)^2\tau^2\left(\int_{t_n}^{t_{n+1}}\|\phi_{tt}\|_0\mathrm{d}s\right)^2+\frac{1}{2}\big\|\xi_\phi^{n+1}+\xi_\phi^n\big\|_0^2$$
$$\leq C(1-\epsilon)^2\tau^3\int_{t_n}^{t_{n+1}}\|\phi_{tt}\|_0^2\mathrm{d}s+\big\|\xi_\phi^{n+1}\big\|_0^2+\|\xi_\phi^n\|_0^2. \tag{5.9}$$

Using (3.6), (3.12), (3.13), the Cauchy-Schwarz inequality and triangle inequality, we have

$$\begin{aligned}
|I_3| &= \left|\left(\nabla w^{n+1/2}-\frac{\nabla P_h w^{n+1}+\nabla P_h w^n}{\tau},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right)\right|\\
&\leq C\tau\int_{t_n}^{t_{n+1}}\|\nabla w_{tt}\|_0\mathrm{d}s\big\|\nabla(\xi_\phi^{n+1}+\xi_\phi^n)\big\|_0\\
&\leq C\tau^2\left(\int_{t_n}^{t_{n+1}}\|\nabla w_{tt}\|_0\mathrm{d}s\right)^2+\big\|\nabla(\xi_\phi^{n+1}+\xi_\phi^n)\big\|_0^2\\
&\leq C\tau^3\int_{t_n}^{t_{n+1}}\|\nabla w_{tt}\|_0^2\mathrm{d}s+\big\|\xi_\phi^{n+1}+\xi_\phi^n\big\|_0\big\|\tilde{\xi}_w^{n+1}+\tilde{\xi}_w^n\big\|_0\\
&\leq C\tau^3\int_{t_n}^{t_{n+1}}\|\nabla w_{tt}\|_0^2\mathrm{d}s+\frac{1}{4\delta_1}\big\|\xi_\phi^{n+1}+\xi_\phi^n\big\|_0+2\delta_1\big\|\xi_w^{n+1}+\xi_w^n\big\|_0\\
&\quad+4\delta_1\big\|\eta_w^{n+1}+\eta_w^n\big\|_0+4\delta_1\big\|\tilde{\eta}_w^{n+1}+\tilde{\eta}_w^n\big\|_0.
\end{aligned}\tag{5.10}$$

Similar arguments lead to

$$\begin{aligned}
|I_4| &= 2\left|\left(\nabla\phi^{n+1/2}-\frac{\nabla P_h\phi^{n+1}+\nabla P_h\phi^n}{2},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right)\right|\\
&= 2\left|\left(\nabla\phi^{n+1/2}-\frac{\nabla\phi^{n+1}+\nabla\phi^n}{2},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right)\right|\\
&\leq C\tau\int_{t_n}^{t_{n+1}}\|\nabla\phi_{tt}\|_0\mathrm{d}s\big\|\nabla(\xi_\phi^{n+1}+\xi_\phi^n)\big\|_0\\
&\leq C\tau^2\left(\int_{t_n}^{t_{n+1}}\|\nabla\phi_{tt}\|_0\mathrm{d}s\right)^2+\big\|\nabla(\xi_\phi^{n+1}+\xi_\phi^n)\big\|_0^2\\
&\leq C\tau^3\int_{t_n}^{t_{n+1}}\|w_{tt}\|_0^2\mathrm{d}s+\big\|\xi_\phi^{n+1}+\xi_\phi^n\big\|_0\big\|\tilde{\xi}_w^{n+1}+\tilde{\xi}_w^n\big\|_0\\
&\leq C\tau^3\int_{t_n}^{t_{n+1}}\|w_{tt}\|_0^2\mathrm{d}s+\frac{1}{4\delta_2}\big\|\xi_\phi^{n+1}+\xi_\phi^n\big\|_0+2\delta_2\big\|\xi_w^{n+1}+\xi_w^n\big\|_0\\
&\quad+4\delta_2\big\|\eta_w^{n+1}+\eta_w^n\big\|_0+4\delta_2\big\|\tilde{\eta}_w^{n+1}+\tilde{\eta}_w^n\big\|_0.
\end{aligned}\tag{5.11}$$

Now we have to analyze the term I_5, which has the following decomposition:

$$I_5=I_{51}+I_{52}, \tag{5.12}$$

where

$$I_{51} = \left(|\phi^{n+1/2}|^2\phi^{n+1/2}, \xi_\phi^{n+1} + \xi_\phi^n\right) - \left(\frac{(|\phi^{n+1}|^2 + |\phi^n|^2)(\phi^{n+1} + \phi^n)}{4}, \xi_\phi^{n+1} + \xi_\phi^n\right), \tag{5.13}$$

$$I_{52} = \left(\frac{(|\phi^{n+1}|^2 + |\phi^n|^2)(\phi^{n+1} + \phi^n)}{4}, \xi_\phi^{n+1} + \xi_\phi^n\right) - \left(\frac{(|P_h\phi^{n+1}|^2 + |P_h\phi^n|^2)(P_h\phi^{n+1} + P_h\phi^n)}{4}, \xi_\phi^{n+1} + \xi_\phi^n\right). \tag{5.14}$$

It follows from Taylor expansion, the Cauchy-Schwarz inequality, triangle inequality and regularity results in Lemma 2.2 that

$$\begin{aligned}
|I_{51}| &\le \left|\left(|\phi^{n+1/2}|^2\left(\phi^{n+1/2} - \frac{\phi^{n+1} + \phi^n}{2}\right), \xi_\phi^{n+1} + \xi_\phi^n\right)\right| \\
&\quad - \left|\left(\left(|\phi^{n+1/2}|^2 - \frac{(|\phi^{n+1}|^2 + |\phi^n|^2)}{2}\right)\frac{(\phi^{n+1} + \phi^n)}{2}, \xi_\phi^{n+1} + \xi_\phi^n\right)\right| \\
&\le C\tau\int_{t_n}^{t_{n+1}} \|\phi_{tt}\|_0 \mathrm{d}s \left\|\xi_\phi^{n+1} + \xi_\phi^n\right\|_0 \\
&\le C\tau^2\left(\int_{t_n}^{t_{n+1}} \|\phi_{tt}\|_0 \mathrm{d}s\right)^2 + \frac{1}{2}\left\|\xi_\phi^{n+1} + \xi_\phi^n\right\|_0^2 \\
&\le C\tau^3\int_{t_n}^{t_{n+1}} \|\phi_{tt}\|_0^2 \mathrm{d}s + \left\|\xi_\phi^{n+1}\right\|_0^2 + \|\xi_\phi^n\|_0^2.
\end{aligned} \tag{5.15}$$

Similarly, using (3.18) yields

$$\begin{aligned}
|I_{52}| &\le C\left\|\eta_\phi^{n+1} + \eta_\phi^n\right\|_0 \left\|\xi_\phi^{n+1} + \xi_\phi^n\right\|_0 \\
&\le C\left\|\eta_\phi^{n+1} + \eta_\phi^n\right\|_0^2 + \frac{1}{2}\left\|\xi_\phi^{n+1} + \xi_\phi^n\right\|_0^2 \\
&\le C\left\|\eta_\phi^{n+1} + \eta_\phi^n\right\|_0^2 + \left\|\xi_\phi^{n+1}\right\|_0^2 + \|\xi_\phi^n\|_0^2.
\end{aligned} \tag{5.16}$$

Combining (5.12)-(5.16) gives

$$|I_5| \le C\tau^3\int_{t_n}^{t_{n+1}} \|\phi_{tt}\|_0^2 \mathrm{d}s + C\left\|\eta_\phi^{n+1} + \eta_\phi^n\right\|_0^2 + \left\|\xi_\phi^{n+1}\right\|_0^2 + \|\xi_\phi^n\|_0^2. \tag{5.17}$$

Similar to (3.18), we assume the nonlinear term in the left-hand side of (5.7) is non-negative

$$0 \le \left(\frac{(|\phi_h^{n+1}|^2 + |\phi_h^n|^2)(\phi_h^{n+1} + \phi_h^n)}{4}, \xi_\phi^{n+1} + \xi_\phi^n\right) - \left(\frac{(|P_h\phi^{n+1}|^2 + |P_h\phi^n|^2)(P_h\phi^{n+1} + P_h\phi^n)}{4}, \xi_\phi^{n+1} + \xi_\phi^n\right).$$

The same argument as (3.13) leads to

$$\begin{aligned}&\left(\frac{\nabla\xi_w^{n+1}+\nabla\xi_w^n}{2},\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\right)\\&=\frac{1}{2}\|\xi_w^{n+1}+\xi_w^n\|_0^2+\frac{1}{2}\big(\xi_w^{n+1}+\xi_w^n,(\eta_w^{n+1}+\eta_w^n)-(\tilde\eta_w^{n+1}+\tilde\eta_w^n)\big)\\&\geq\left(\frac{1}{2}-2\delta_3\right)\|\xi_w^{n+1}+\xi_w^n\|_0^2-C\|\eta_w^{n+1}+\eta_w^n\|_0^2-C\|\tilde\eta_w^{n+1}+\tilde\eta_w^n\|_0^2.\end{aligned}$$

From (3.14), we have

$$\begin{aligned}&\frac{1-\epsilon}{2}\big(\tilde\xi_\phi^{n+1}+\tilde\xi_\phi^n,\xi_\phi^{n+1}+\xi_\phi^n\big)\\&\leq\frac{3(1-\epsilon)}{4}\|\xi_\phi^{n+1}+\xi_\phi^n\|_0^2+\frac{1-\epsilon}{4}\|\eta_\phi^{n+1}+\eta_\phi^n\|_0^2.\end{aligned}$$

The combination of (3.12), (3.15) and the Cauchy-Schwarz inequality leads to

$$\begin{aligned}&2\big(\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n,\nabla\xi_\phi^{n+1}+\nabla\xi_\phi^n\big)\\&=2\big(\tilde\xi_w^{n+1}+\tilde\xi_w^n,\xi_\phi^{n+1}+\xi_\phi^n\big)\\&\leq2\delta_4\|\xi_w^{n+1}+\xi_w^n\|_0^2+\left(1+\frac{1}{2\delta_4}\right)\|\xi_\phi^{n+1}+\xi_\phi^n\|_0^2+\|\eta_w^{n+1}+\eta_w^n\|_0^2.\end{aligned}$$

We choose $\delta_1,\delta_2,\delta_3$ and δ_4 such that

$$2(\delta_1+\delta_2+\delta_3+\delta_4)=\frac{1}{4},$$

e.g., $\delta_1=\delta_2=\delta_3=\delta_4=1/32$. Thus, combining (5.7), (5.8)-(5.11) and (5.17), we have

$$\begin{aligned}&\left(\frac{\xi_\phi^{n+1}-\xi_\phi^n}{\tau},\xi_\phi^{n+1}+\xi_\phi^n\right)+\frac{1}{4}\|\xi_w^{n+1}+\xi_w^n\|_0^2\\&\leq C\tau^3\int_{t_n}^{t_{n+1}}\|\phi_{ttt}\|_0^2\mathrm{d}s+C(1-\epsilon)^2\tau^3\int_{t_n}^{t_{n+1}}\|\phi_{ttt}\|_0^2\mathrm{d}s+C\tau^3\int_{t_n}^{t_{n+1}}\|\phi_{tt}\|_0^2\mathrm{d}s\\&\quad+C\tau^3\int_{t_n}^{t_{n+1}}\|\nabla w_{tt}\|_0^2\mathrm{d}s+\frac{C}{\tau}\int_{t_n}^{t_{n+1}}\|\partial_t\eta_\phi\|_0^2\mathrm{d}s+\tilde C\big(\|\xi_\phi^{n+1}\|_0^2+\|\xi_\phi^n\|_0^2\big)\\&\quad+C\|\eta_w^{n+1}+\eta_w^n\|_0^2+C\|\tilde\eta_w^{n+1}+\tilde\eta_w^n\|_0^2+C\|\eta_\phi^{n+1}+\eta_\phi^n\|_0^2.\end{aligned}$$

Summing both sides with respect to n gives

$$\begin{aligned}&\|\xi_\phi^N\|_0^2-\|\xi_\phi^0\|_0^2+\frac{1}{4}\sum_{n=0}^{N-1}\tau\|\xi_w^{n+1}+\xi_w^n\|_0^2\\&\leq C\tau^4\int_0^T\|\phi_{ttt}\|_0^2\mathrm{d}s+C(1-\epsilon)^2\tau^4\int_0^T\|\phi_{ttt}\|_0^2\mathrm{d}s+C\tau^4\int_0^T\|\phi_{tt}\|_0^2\mathrm{d}s\end{aligned}$$

$$+C\tau^4\int_0^T\|\nabla w_{tt}\|_0^2\mathrm{d}s+C\int_0^T\|\partial_t\eta_\phi\|_0^2\mathrm{d}s+C\sum_{n=0}^{N-1}\tau\big\|\eta_w^{n+1}+\eta_w^n\big\|_0^2$$
$$+C\sum_{n=0}^{N-1}\tau\big\|\tilde{\eta}_w^{n+1}+\tilde{\eta}_w^n\big\|_0^2+C\sum_{n=0}^{N-1}\tau\big\|\eta_\phi^{n+1}+\eta_\phi^n\big\|_0^2+2\tilde{C}\tau\sum_{n=0}^{N}\|\xi_\phi^n\|_0^2.$$

Then Gronwall's inequality in Lemma 4.1 yields the desired result (5.6), provided that $2\tilde{C}\tau<1$. □

Proof of Theorem 5.1. The desired result (5.1) can be established through a reasoning process analogous to that used in Theorem 3.2, supplemented by the application of the triangle inequality. □

6. Numerical examples

In this section, two numerical examples are presented to illustrate the efficiency of our mixed finite element method and the CN scheme (4.1) for the SH equation (2.4).

Example 6.1. In the first example, we test the convergence rate stated in Theorem 5.1. For the aim of testing the convergence order, we consider the following SH problem:

$$\begin{cases}\partial_t\phi=-\Delta^2\phi-2\Delta\phi-\phi^3-(1-\epsilon)\phi+f & \text{in } \Omega\times(0,T),\\ \phi(\mathbf{x},0)=\phi_0(\mathbf{x}) & \text{in } \Omega,\end{cases}$$

where $\Omega=(0,2)\times(0,2), T=1$, 2-periodic boundary condition, and $\epsilon=0.01$. We introduce the source term f in order to make the exact solution known. Here the initial solution $\phi_0(\mathbf{x})$ and the function f are chosen such that the exact solution is

$$\phi(\mathbf{x},t)=0.1e^{-t}\sin(\pi x_1)\sin(\pi x_2).$$

We use the following linear and quadratic elements:

$$V_h:=\left\{v_h\in H^1_{\text{per}}(\Omega),\, v_h|_K\in\mathcal{P}_1(K)\ \text{ or }\ \mathcal{P}_2(K),\ \forall K\in\mathcal{T}_h\right\}.$$

In order to test the convergence rate, we take the following error:

$$\sqrt{\|\phi(\mathbf{x},T)-\phi_h^N\|_0^2+\frac{1}{4}\sum_{n=0}^{N}\tau\|w(\mathbf{x},t_n)-w_h^n\|_0^2}.$$

The numerical results are illustrated in Fig. 1. In the left subfigure, it is evident that the convergence rate for the linear finite element method exceeds by one order the predictions made in Theorem 5.1, a discrepancy that warrants further investigation. Additionally, we examined the temporal convergence order of scheme (4.1) in this example. As shown in the right subfigure, our method exhibits second-order accuracy in time, which aligns with the findings presented in Theorem 5.1.

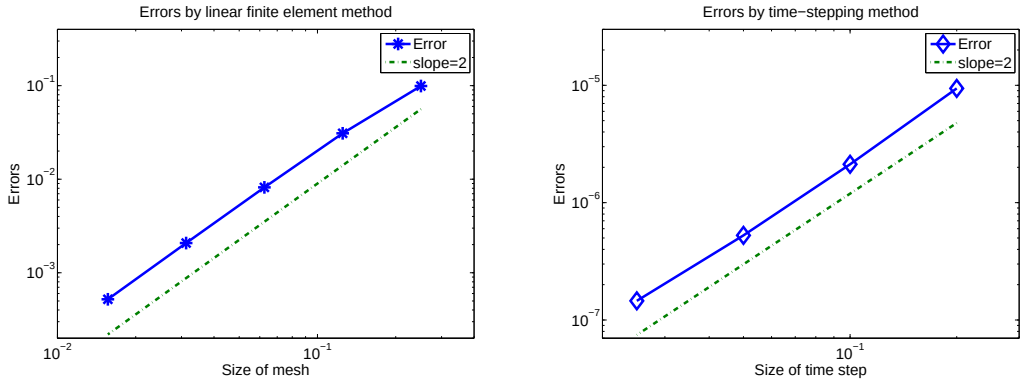


Figure 1: The convergence behavior of the linear finite element method with fixed time step size $\tau = 0.01$ (left), and the time-stepping method with the quadratic finite element and fixed spatial size $h = 1/64$ (right).

Example 6.2. In the second example, we discuss the following SH problem:

$$\begin{cases} \partial_t \phi = -\Delta^2 \phi - 2\Delta\phi - \phi^3 - (1-\epsilon)\phi & \text{in } \Omega \times (0,T), \\ \phi(\mathbf{x},0) = \phi_0(\mathbf{x}) & \text{in } \Omega, \end{cases}$$

where $\Omega = (0,32) \times (0,32)$, 32-periodic boundary condition, $\epsilon = 0.025$. And the initial data is chosen as

$$\begin{aligned} \phi_0(\mathbf{x}) = 0.07 &- 0.02\cos\left(\frac{2\pi(x_1-12)}{32}\right)\sin\left(\frac{2\pi(x_2-1)}{32}\right) \\ &+ 0.02\cos^2\left(\frac{\pi(x_1+10)}{32}\right)\cos^2\left(\frac{\pi(x_2+3)}{32}\right) \\ &- 0.01\sin^2\left(\frac{4\pi x_1}{32}\right)\sin^2\left(\frac{4\pi(x_2-6)}{32}\right). \end{aligned}$$

In this example, we apply a mixed linear finite element method combined with the CN time-stepping scheme (4.1) to address the problem. Fig. 2 illustrates the solution contours at various time points: $t = 0$, $t = 0.5$, $t = 2$, $t = 5$, $t = 10$, and $t = 50$. This approach allows us to visualize the evolution of the solution over time, providing insights into the dynamic behavior of the system at these specific instances.

Additionally, we examined the energy decay property of this problem. Fig. 3 presents the numerical result, and we find the expected energy decay behavior. This analysis confirms that the system exhibits the anticipated reduction in energy over time, validating the theoretical expectations.

7. Concluding remarks

In this paper, we have devised and analyzed a mixed finite element method for solving the Swift-Hohenberg equation. By decomposing the biharmonic operator within the

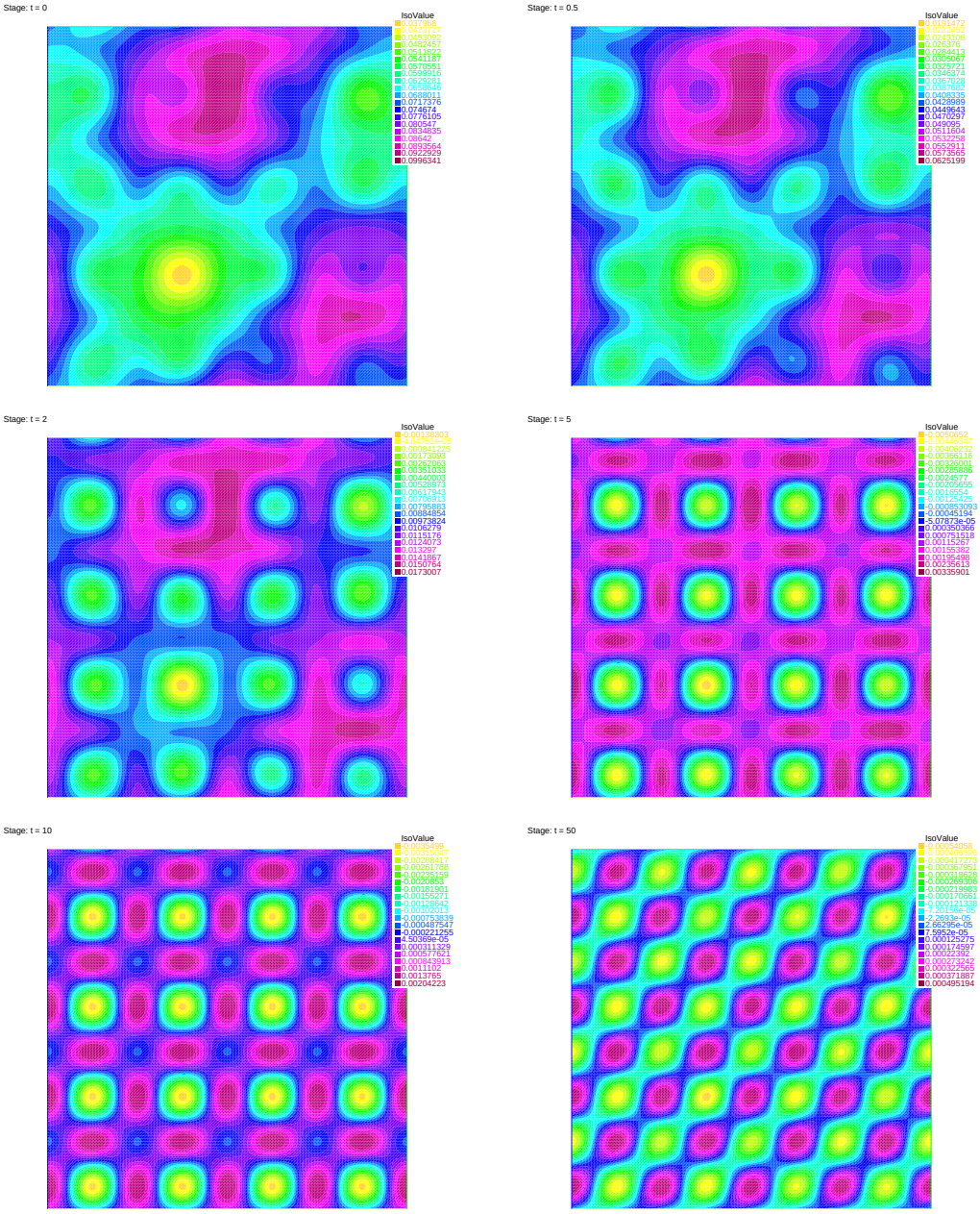


Figure 2: The solution contours at different time.

Swift-Hohenberg models into a sequence of two Laplacian operators, we were able to employ standard continuous finite element spaces that are commonly utilized for elliptic problems. Our focus then shifted to exploring the gradient flow characteristics of the resulting numerical schemes, where we established discrete analogues of the energy decay properties observed in the continuous case. Through meticulous energy

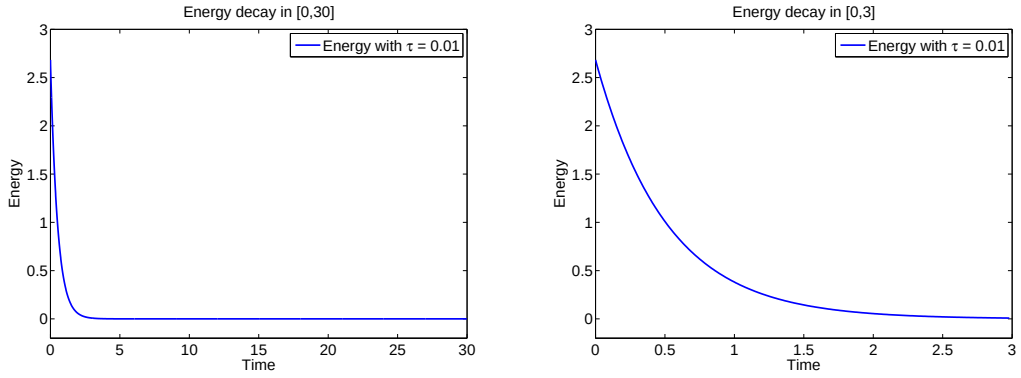


Figure 3: The energy decay with $\tau = 0.01$ and $h = 1/4$.

estimations, we confirmed the fully stable property, as stated in Theorem 4.2. Building upon these stability findings, we derived corresponding error estimates. Ultimately, the theoretical predictions concerning stability and convergence were validated through two numerical experiments, thereby corroborating our analytical results.

Acknowledgments

The first author was supported in part by the Project of Social Science Foundation of Jiangsu Province (Grant No. 22JYD001). The second author was supported in part by the National Natural Science Foundation of China (Grant Nos. 12001402, 12571437). The third author was supported in part by the National Natural Science Foundation of China (Grant No. 12201017) and by the Disciplinary funding of Beijing Technology and Business University.

References

[1] R. A. ADAMS, *Sobolev Spaces*, Academic Press, 1975.
[2] H. BRUNNER, *Collocation Methods for Volterra Integral and Related Functional Differential Equations*, Cambridge University Press, 2004.
[3] P. G. CIARLET AND J. L. LIONS (EDS.), *Finite Element Methods*, Handbook of Numerical Analysis, Vol. 2, North-Holland, 1991.
[4] M. CROSS AND P. HOHENBERG, *Pattern formation outside of equilibrium*, Rev. Mod. Phys. 65(3) (1993), 851.
[5] K. R. ELDER, M. KATAKOWSKI, M. HAATAJA, AND M. GRANT, *Modeling elasticity in crystal growth*, Phys. Rev. Lett. 88 (2002), 245701.
[6] R. GABBRIELLI, *A new counter-example to Kelvin's conjecture on minimal surfaces*, Phil. Mag. Lett. 89(8) (2009), 483–491.
[7] R. GUO AND Y. XU, *Local discontinuous Galerkin method and high order semi-implicit scheme for the phase field crystal equation*, SIAM J. Sci. Comput. 38(1) (2016), A105–A127.

[8] Z. Hu, S. M. Wise, C. Wang, and J. S. Lowengrub, *Stable and efficient finite-difference nonlinear-multigrid schemes for the phase field crystal equation*, J. Comput. Phys. 228(15) (2009), 5323–5339.
[9] A. Hutt and F. Atay, *Analysis of nonlocal neural fields for both general and gamma-distributed connectivities*, Physica D 203 (2005), 30–54.
[10] S. Kesavan, *Topics in Functional Analysis and Applications*, Wiley Eastern Limited, 1989.
[11] H. G. Lee, *A semi-analytical Fourier spectral method for the Swift-Hohenberg equation*, Comput. Math. Appl. 74 (2017), 1885–1896.
[12] J. Lega, J. Moloney, and A. Newell, *Swift-Hohenberg equation for lasers*, Phys. Rev. Lett. 73 (1994), 2978–2981.
[13] B. Li and J. G. Liu, *Thin film epitaxy with or without slope selection*, European J. Appl. Math. 14 (2003), 713–743.
[14] Y. Li and J. Kim, *An efficient and stable compact fourth-order finite difference scheme for the phase field crystal equation*, Comput. Methods. Appl. Mech. Eng. 319 (2017), 194–216.
[15] H. Liao, B. Ji, and L. Zhang, *An adaptive BDF2 implicit time-stepping method for the phase field crystal model*, IMA J. Numer. Anal. 42(1) (2022), 649–679.
[16] H. Liu and P. Yin, *Unconditionally energy stable DG schemes for the Swift-Hohenberg equation*, J. Sci. Comput. 81 (2019), 789–819.
[17] Z. Liu, *Novel energy stable schemes for Swift-Hohenberg model with quadratic-cubic nonlinearity based on the H^{-1}-gradient flow approach*, Numer. Algorithms 87(2) (2021), 633–650.
[18] Z. Liu and C. Chen, *On efficient semi-implicit auxiliary variable methods for the six-order Swift-Hohenberg model*, J. Comput. Appl. Math. 419 (2023), 114730.
[19] L. Qi and Y. Hou, *A second order energy stable BDF numerical scheme for the Swift-Hohenberg equation*, J. Sci. Comput. 88 (2021), 1–25.
[20] L. Qi and Y. Hou, *Error analysis of first- and second-order linear, unconditionally energy-stable schemes for the Swift-Hohenberg equation*, Comput. Math. Appl. 127 (2022), 192–212.
[21] Z. Qiao, T. Tang, and H. Xie, *Error analysis of a mixed finite element method for the molecular beam epitaxy model*, SIAM J. Numer. Anal. 53(1) (2015), 184–205.
[22] J. Shen, J. Xu, and J. Yang, *The scalar auxiliary variable (SAV) approach for gradient flows*, J. Comput. Phys. 353 (2017), 407–416.
[23] J. Swift and P. Hohenberg, *Hydrodynamic fluctuations at the convective instability*, Phys. Rev. A 15 (1977), 319–328.
[24] L. Wang and Y. Huang, *Error estimates for second-order SAV finite element method to phase field crystal model*, Electron. Res. Arch. 29(1) (2021), 1735–1752.
[25] M. Wang, Q. Huang, and C. Wang, *A second order accurate scalar auxiliary variable (SAV) numerical method for the square phase field crystal equation*, J. Sci. Comput. 88 (2021), 33.
[26] S. M. Wise, C. Wang, and J. S. Lowengrub, *An energy-stable and convergent finite-difference scheme for the phase field crystal equation*, SIAM J. Numer. Anal. 47(3) (2009), 2269–2288.
[27] J. Yang, Z. Tan, and J. Kim, *High-order time-accurate, efficient, and structure-preserving numerical methods for the conservative Swift-Hohenberg model*, Comput. Math. Appl. 102 (2021), 160–174.
[28] X. Yang and D. Han, *Linearly first- and second-order, unconditionally energy stable schemes for the phase field crystal model*, J. Comput. Phys. 330 (2017), 1116–1134.
[29] Z. Zhang and Y. Ma, *On a large time-stepping method for the Swift-Hohenberg equation*,

Adv. Appl.Math. Mech. 8 (2016), 992–1003.

[30] L. ZHOU AND R. GUO, *Optimal error estimates of the local discontinuous Galerkin method and high-order time discretization scheme for the Swift-Hohenberg equation*, J. Sci. Comput. 93(2) (2022), 46.